Supplementary Material:
Majorana zero modes in silicon

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Contents

1 Model, basis, and the physical pairing channel

1.1 Single-channel BdG conventions

The normal-state Hamiltonian on N sites with spacing dx is

$$h = \sum_n c_n^\dagger[(2t - \mu + V_n)\sigma_0 + E_Z\sigma_x]c_n + \sum_n \left[c_{n+1}^\dagger(-t\sigma_0 - i\, a_{so}\sigma_z)c_n + \text{h.c.}\right], \tag{1}$$

with $t = \hbar^2/2m^*dx^2$, $a_{so} = \alpha/2dx$, $E_Z = \frac{1}{2}g\mu_B B$, and singlet pairing $\Delta\, i\sigma_y$. The BdG matrix is assembled as $H = \begin{pmatrix} h & D \\ D^\dagger & -h^* \end{pmatrix}$, $D = \Delta \nVdash_N \otimes i\sigma_y$, which enforces particle–hole symmetry $\mathcal{P} = \tau_x K$

exactly on the lattice. The topological criterion at μ measured from the band bottom is $E_Z^2 > \Delta^2 + \mu^2$. All energies are handled in SI units (J) internally and quoted in μeV.

1.2 Two-valley basis and why pairing is inter-valley

The conduction-band wire has two relevant low-energy valleys at $\pm k_0 \hat{z}$, $k_0 = 0.85\,(2\pi/a_{\mathrm{Si}})$. In the envelope basis (c_+, c_-) (valley Pauli matrices ν_i), time reversal maps $+k_0 \uparrow \to \; -k_0 \downarrow$: *the two valleys are time-reversed partners.* A uniform s-wave parent induces pairing between time-reversed states at zero total momentum; the only zero-momentum singlet channel is therefore *inter*-valley,

$$D \;=\; \Delta\,(\nu_x \otimes i\sigma_y), \qquad D^T = -D. \tag{2}$$

Intra-valley pairing (c_+c_+) would carry total momentum $2k_0$ (an FFLO-type condensate) and is not induced by a uniform parent; it is kept only in the legacy fig. 4/6 toy model for comparison. Valley splitting enters as valley–orbit (VO) coupling $\lambda(x) = |\lambda(x)|e^{i\phi(x)}$,

$$h_{VO}(x) = |\lambda(x)|\,[\cos\phi(x)\,\nu_x + \sin\phi(x)\,\nu_y] \otimes \sigma_0, \tag{3}$$

with splitting $E_v = 2|\lambda|$; a perfect interface has constant λ, and the phase ϕ is set by the interface position (Sec. 3).

2 The valley equivalence theorem

Theorem 1 (smooth-interface equivalence)**.** *For any* uniform *VO phase ϕ and amplitude $|\lambda|$, the two-valley wire with inter-valley pairing (2), arbitrary valley-scalar terms (kinetic, μ, $V(x)$, Zeeman), and either valley-scalar Rashba SOC ($\alpha k\,\nu_0\sigma_z$) or phase-locked Dresselhaus SOC ($\alpha k[\cos\phi\,\nu_x + \sin\phi\,\nu_y]\sigma_z$) is unitarily equivalent to two* decoupled *single-band wires with chemical potentials $\mu\mp|\lambda|$, pairing $\pm\Delta$, and the full SOC strength $\pm\alpha$ in each band.*

Proof. Let $U_\phi = e^{-i\phi\nu_z/2} \otimes \sigma_0$ act identically on every site. (i) Valley-scalar terms commute with U_ϕ. (ii) The VO term rotates to $|\lambda|\,\nu_x\sigma_0$ since $U_\phi(\cos\phi\,\nu_x + \sin\phi\,\nu_y)U_\phi^\dagger = \nu_x$. (iii) The pairing block transforms as $D \to U_\phi D\,U_\phi^T$ (not $U^\dagger$ on the right: D couples two creation operators). Because $\nu_z^T = \nu_z$ we have $U_\phi^T = U_\phi$, and using $\nu_x e^{-i\phi\nu_z/2} = e^{+i\phi\nu_z/2}\nu_x$,

$$U_\phi\,(\Delta\nu_x i\sigma_y)\,U_\phi^T = \Delta\,e^{-i\phi\nu_z/2}\nu_x e^{-i\phi\nu_z/2}\,i\sigma_y = \Delta\,\nu_x\,i\sigma_y. \tag{4}$$

The inter-valley pairing is exactly invariant under the valley rotation, for any ϕ. (iv) The phase-locked Dresselhaus SOC rotates, by the same algebra as (ii), to $\alpha k\,\nu_x\sigma_z$. After the rotation everything is diagonal in the ν_x eigenbasis $|\pm\rangle = (|+\rangle_v \pm |-\rangle_v)/\sqrt{2}$: the VO term becomes $\pm|\lambda|$ (a chemical-potential shift $\mu\mp|\lambda|$), the pairing $\Delta\nu_x i\sigma_y \to \pm\Delta\,i\sigma_y$, and the Dresselhaus SOC $\to \pm\alpha k\sigma_z$. The sign of Δ is removed per band by $c \to e^{i\pi/2}c$, and the sign of α by $\sigma_z \to -\sigma_z$ (a spin rotation that leaves $E_Z\sigma_x$ invariant only combined with $\sigma_x \to \sigma_x$; explicitly, the unitary σ_x does this). Hence two decoupled standard wires at $\mu \mp |\lambda|$. □

Numerical verification: `tests.py` (check 3) compares the full $8N$ spectrum against the decoupled-band reference for five phases $\phi \in \{0, 0.7, \pi/2, 2.6, 5.0\}$ and both SOC classes, enforcing agreement to $10^{-6}\,\mu$eV on every eigenvalue (spot checks reach $\sim 3\times10^{-10}\,\mu$eV, i.e. solver precision). The practical consequence is the wedge picture of Fig. 4 of the main text: at $E_v \gtrsim 100\,\mu$eV one band can be placed in its topological window while the other is trivial, and a smooth interface is benign *regardless of the VO phase value.*

3 Atomic steps

3.1 The step phase $\Delta\varphi = 2k_0(a/4)$

The VO coupling is generated by the sharp interface potential $U_{\rm int}(z)$, which scatters $+k_0 \to -k_0$ with momentum transfer $2k_0$:

$$\lambda \;\propto\; \int dz\, F^2(z)\, U_{\rm int}(z - z_0)\, e^{2ik_0 z} \;\Longrightarrow\; \arg\lambda \;\supset\; 2k_0 z_0, \tag{5}$$

where $F(z)$ is the smooth subband envelope and z_0 the interface height. A single-atomic step on a (001) surface changes z_0 by $a/4$ ($a = 5.431$ Å; the four inequivalent (001) atomic planes per cubic cell are spaced $a/4$). Hence

$$\Delta\varphi = 2k_0\, \frac{a}{4} = 2 \cdot 0.85 \cdot \frac{2\pi}{a} \cdot \frac{a}{4} = 0.85\,\pi, \tag{6}$$

accidentally close to π because k_0 sits at 85% of the zone boundary in silicon. A double-height (bunched) step gives $2k_0(a/2) = 1.7\pi \equiv -0.3\pi \pmod{2\pi}$, which is *milder* — the basis of the step-bunching design rule in the main text. This is the standard interface-position dependence of valley-orbit coupling [Hosseinkhani–Burkard, PRB 100, 125309 (2019); Losert *et al.*, PRB 108, 125405 (2023)]; what is new here is its consequence under *inter-valley pairing.*

3.2 A step is a Josephson junction: band-projected pairing

Diagonalize the local VO term at fixed x: eigenvectors $|\pm\rangle_\phi = (|+\rangle_v \pm e^{i\phi}|-\rangle_v)/\sqrt{2}$ with energies $\pm|\lambda|$. Projecting the pairing (2) onto the local bands (pairing projects with the *conjugate* right basis, $D_{\rm band} = W^\dagger D W^*$, $W = (|+\rangle_\phi, |-\rangle_\phi)$):

$$\Big(W^\dagger \nu_x W^*\Big)_{\eta\eta'} = \begin{pmatrix} e^{-i\phi} & 0 \\ 0 & -e^{-i\phi} \end{pmatrix}_{\eta\eta'} \;\Longrightarrow\; D_{\rm band} = \pm\,\Delta\, e^{-i\phi(x)}\, i\sigma_y. \tag{7}$$

Each band therefore sees a superconducting order parameter whose phase is minus the local VO phase. A VO phase step $\Delta\varphi$ at x_s is exactly a Josephson junction with phase drop $\Delta\varphi$ in the effective single-band topological wire. The gauge-invariant statement: the relative phase between $\lambda(x)$ and the (uniform) inter-valley pairing cannot be removed — a position-dependent rotation $U_{\phi(x)}$ leaves D invariant [Eq. (4) holds locally] and makes λ uniform, but generates a kinetic gauge field $k \to k - \frac{1}{2}(\partial_x\phi)\nu_z$, i.e. a δ-spike of ν_z-flux at the step; in the band basis ν_z is purely off-diagonal, so the spike both couples the bands and, within each band, reproduces the junction. The two pictures are related by the gauge choice; the spectrum is identical.

At $\Delta\varphi = \pi$ the order parameter changes sign: a topological wire with a π-junction hosts a Kitaev domain wall — an exact zero mode (numerically $E = 0.013\,\mu$eV, Table 1). Near $\Delta\varphi = 0.85\pi$ the bound state is deep. For a transparent short junction the Andreev bound state lies at $E_{ABS} = \Delta_t|\cos(\Delta\varphi/2)|$ with Δ_t the topological gap; Table 1 compares this zero-parameter formula with the lattice numerics.

The formula tracks the numerics to $\lesssim 15\%$ for $\Delta\varphi \lesssim 0.5\pi$ and overestimates by 25–30% (relative to the lattice value) near 0.85π — the expected accuracy for an atomically abrupt junction (zero length but not in the Andreev limit $\Delta \ll$ all normal scales, and the bound state hybridizes with the wire-end MZMs at finite L). We therefore present the junction mechanism as an *analytically derived mapping* [Eq. (7), exact] with a *numerically quantified* bound-state depth, not as a closed-form spectrum.

Table 1: Single-step bound state vs the short-junction Andreev formula ($\Delta_t = 22.22\,\mu$eV, wedge point of Fig. 9; lattice values from `output/data/key_numbers.json`).

$\Delta\varphi/\pi$	0	0.10	0.25	0.40	0.50	0.65	**0.85**	0.95	1.00
lattice E_2 (μeV)	22.22	22.07	19.77	16.08	13.45	9.42	**4.03**	1.34	0.01
$\Delta_t\lvert\cos\frac{\Delta\varphi}{2}\rvert$	22.22	21.95	20.53	17.98	15.71	11.61	**5.19**	1.74	0.00

3.3 Why net winding is irrelevant at realistic step densities

A same-sign staircase accumulates winding $\sum_s \Delta\varphi_s$; a sign-randomized control with identical $|\Delta\varphi|$ accumulates none. Fig. 9(a) of the main text shows the two ensembles are statistically indistinguishable: the damage is per-junction. Winding *does* matter in the smooth limit: a linear ramp $\phi(x) = qx$ is a uniform gauge field, pair-breaking only beyond $q \gtrsim 2|\lambda|/(\hbar v_F)$-type thresholds (in-code generator, `fig9` key numbers: gap reduced by $\lesssim 25\%$ with no subgap state at $q \le 4\times10^6\,\mathrm{m}^{-1}$, destroyed by 1.6×10^7); at 50 nm step spacing the equivalent $q = 0.85\pi/50\,\mathrm{nm} = 5.3\times10^7$ exceeds this, but the staircase damage at the same mean gradient is dominated by the discrete junction physics, as the control comparison proves.

4 Complete parameter tables

Baseline silicon parameters used throughout unless a figure row overrides them: $m^* = 0.19\,m_e$, $g = 2.0$, $\Delta = 50\,\mu$eV (induced), $dx = 5\,$nm, $\alpha_{\text{demo}} = 0.05\,$eV Å (the "engineered" demonstration value), $\alpha_{\text{Si,opt}} = 10^{-3}\,$eV Å, $\alpha_{\text{InSb}} = 0.5\,$eV Å. Disorder is onsite iid uniform in $[-W, W]$ with correlation length $= dx$ (thresholds are lattice-convention dependent and quoted with dx). Bulk gaps minimize the lower BdG band over $n_k = 6001$ points unless noted. Seeds are `numpy default_rng` with the listed integer families; every random draw in the repository is seeded.

Table 2: Electron-sector figures. $L = N\,dx$. "IV" = inter-valley pairing model [Eq. (2)]; "toy" = legacy intra-valley pairing.

Fig.	model	N (dx)	μ (μeV)	B (T)	α (eV Å)	valley	disorder	notes
1	1-band	400 (5 nm)	0	0.02–2 (40)	0.05	—	—	validation: fan, MZM, splitting
2	bulk	n_k=6001	−150..150 (201)	0.01–3 (181)	10^{-3} / 0.5	—	—	phase diagrams
3	bulk opt	n_μ=16, n_B=18	0–145	≤ 2	10^{-4}..0.5	—	—	$\Delta(B)$ GL optional
4	toy	200 (5 nm)	−150..150 (31)	1.5	0.05	E_v: 0–300 (31), δ_{iv}=10	—	seed [4-family]
5	1-band	L/dx (case)	0	1.5	case	—	W per case, 16 seeds	cases below
6	toy vs IV	300 (5 nm)	35	1.5	0.05	E_v: 0–300 (13)	—	8 seeds; steps L_s=30,100 nm
7	2D strip	300×5 (5×10 nm)	subband windows	1.5	0.05	—	—	B along wire
9	IV	500 (5 nm)	35	1.5	0.05	$\lvert\lambda\rvert$=75	steps: 6 spacings, 14 seeds	seed family [71,·]
12	RGF	L=2 μm	0	1.5	0.05	—	$W \in \{100, 200, 400, 800\}$	Ando leads, E-scan

Table 3: Fig. 5 disorder cases (per-case dx, L chosen for $k_{so}dx < 0.1$ and $L \gg \xi$ where affordable; 16 realizations each).

case	α (eV Å)	dx, L	W grid (μeV)
intrinsic Si	0.001	5 nm, 20 μm	0, 5, 10, 20, 50, 100, 200, 400
engineered Si	0.05	2.5 nm, 2 μm	0, 20, 50, 100, 200, 400, 800, 1600
strong SOC	0.15	2.5 nm, 6 μm	0, 20, 50, 100, 200, 400, 800, 1600

Table 4: Hole-sector figures. Parent models in Sec. 6. Optimizer: gap maximized over $B \leq B_{\max}(n_B{=}7)$, $\Delta_{ind} \in [8, \Delta_p(B)]$ ($n_\Delta{=}7$), $\mu \in [0, 160]\,\mu$eV ($n_\mu{=}14$), $n_k{=}2501$ (convergence: Sec. 5).

Fig.	model	parameters
8	bulk opt	(α, g) box: $\alpha \in [0.01, 0.3]$ eV Å (log, 20), $g \in [1.2, 4.0]$ (20); $m^* = 0.25$; center point $(\alpha, g) = (0.06, 2.2)$; parents Si:B-Pauli / Si:B-measured / Al; metallization $Z = 1 - \Delta_{ind}/\Delta_p$, $\alpha \to Z\alpha$, $g \to Zg + (1{-}Z)\, g_{\text{parent}}$, $g_{\text{parent}} = 2$. Finite-wire check at center: $N{=}1200$, $dx{=}2.5$ nm, 10 seeds, $W \leq 800\,\mu$eV.
10	4-band LK	fin $W_y{\times}W_z = 10{\times}12$ nm hard wall, grid $n_y{\times}n_z = 11{\times}13$; $\gamma_1, \gamma_2, \gamma_3 = 4.285, 0.339, 1.446$, $\kappa = -0.42$ (Lawaetz); $E_z \in \{2, 5, 10, 15, 20, 30, 40, 50\}$ MV/m; bare + orbital Zeeman; no strain, no cubic J^3 terms (stated caricature).
11	orientation	$\theta \in [0°, 90°]$ (8), $\phi_B \in [0°, 180°]$ (13); LK tensor at $E_z = 10$ MV/m ($\alpha{=}0.054$, $\lvert g\rvert{=}(0.5\text{–}1.1,\ g_y,\ 2.4\text{–}4.2)$, SOC axis $\hat{y}$) vs empirical tensor ($\alpha{=}0.06$, $g{=}(2.1, 2.35, 2.7)$, SOC axis $0.41\hat{y} + 0.91\hat{z}$); parents: thick Si:B (isotropic GL, $B_{c2}{=}0.4$ T) and Al film ($B_{c\parallel}{=}2$ T, $B_{c\perp}{=}0.1$ T); tilted-gap $n_k{=}500$ with the Osca–Ruiz–Serra criterion.
13	convergence	Sec. 5; engine `convergence.py`.

5 Convergence and sensitivity

All entries regenerate from `convergence.py` (`output/data/convergence.json`); Fig. 1 summarizes. The observables are the clean two-valley wedge gap and the single-step (0.85π) bound state at the Fig. 9 baseline ($\mu = 35\,\mu$eV, $B = 1.5$ T, $E_v = 150\,\mu$eV), plus the hole-platform optimizer outputs.

Table 5: Discretization and length convergence (μeV).

	dx (nm), $L{=}2.5\,\mu$m				L (μm), $dx{=}5$ nm		
	10	**5**	2.5		1.25	**2.5**	5.0
clean gap	22.165	22.221	22.234	clean	21.566	22.221	22.239
step 0.85π	4.037	4.034	4.033	step	4.149	4.034	4.034
π-step E_0	0.013	0.013	0.013	ens. median[a]	2.537	0.648	0.083

[a] 50-nm same-sign Poisson staircase, 14 seeds. The L-dependence is physical (Anderson localization; transport section of the main text): ensemble medians are quoted only as L-tagged bounds.

Table 6: Statistical and parametric sensitivity.

seeds n	7	14	28	56
running median (μeV)	0.724	0.648	0.583	0.638
56-seed bootstrap 95% CI: $[0.516, 0.810]\,\mu$eV				
μ (μeV)	15	25	**35**	45/60
clean / step	9.7 / 1.9	16.7 / 3.0	22.2 / 4.0	24.6–27.6 / 5.0–6.2
E_v (μeV)	100	**150**	225	300
clean / step	11.6 / 2.6	22.2 / 4.0	exits topological window at fixed μ[b]	

[b] At fixed $\mu = 35\,\mu$eV, $B = 1.5$ T the lower band leaves its topological window for $E_v > 212\,\mu$eV (analytic; grid points $\geq 225\,\mu$eV are outside); single-valley topology at larger E_v requires re-centering μ (main-text Fig. 4 wedge). The quoted values at $E_v \geq 225$ are trivial gaps and are excluded from any protection claim.

The one headline number that moves under refinement is the measured-field Si:B center point: $9.9 \to 11.0\,\mu$eV bare ($9.1 \to 9.8$–10.1 renormalized) at finer optimizer grids — the production grid was ~10% *conservative*. The Pauli-scenario and Al numbers are grid-exact. Main-text ranges are

Table 7: Hole-platform optimizer convergence and parent-model sensitivity (gap in μeV; (r) = metallization-renormalized).

grid $n_B \times n_\Delta \times n_\mu$	5×5×10	7×7×14	10×10×20	14×14×28
Si:B measured	8.83	9.92	11.01	11.01
Si:B measured (r)	7.93	9.14	10.13	9.82
Si:B Pauli	30.62	30.62	30.62	30.62
Si:B Pauli (r)	19.68	19.68	19.68	20.21

k-grid n_k: $\leq 0.022\%$ drift from 501 to 12001 at all three operating points.
Parent model: $\Delta_0 \times \{0.8, 1, 1.2\}$, $B_{c2} \in \{0.3, 0.4\}$ T $\Rightarrow$ measured-field center 8.0–10.6 (7.6–10.0 r).

quoted accordingly.

6 Parent superconductor models and the $g = 2$ optimum

6.1 What Δ is

Δ in the BdG Hamiltonian is a *static induced gap*: the infinite-coupling limit of the standard tunneling self-energy $\Sigma(\omega) = -\gamma\,(\omega + \Delta_p \tau_x)/\sqrt{\Delta_p^2 - \omega^2}$ evaluated at $\omega = 0$, with $\Delta_{ind} = \gamma\Delta_p/(\gamma + \Delta_p)$. We do not solve the dynamical problem; instead (i) field suppression enters through the parent, $\Delta_{ind} \leq \Delta_p(B)$, with $\Delta_p(B) = \Delta_0[1 - (B/B_c)^2]$ (GL, orbital-limited parents) or $\Delta_0 - \mu_B B$ (Pauli-limited caricature), and (ii) metallization enters through the quasiparticle weight $Z = 1 - \Delta_{ind}/\Delta_p$, which renormalizes $\alpha \to Z\alpha$ and $g \to Zg + (1-Z)g_p$ ($g_p = 2$). Both caricatures are stated wherever used; replacing them by the full self-energy is listed as future work and would *lower* all gaps further (the renormalized numbers bound the direction).

6.2 Connection to Cole–Das Sarma–Stanescu

Ref. [Cole *et al.*, PRB 92, 174511 (2015)] showed that strong semiconductor–superconductor coupling buys induced-gap size at the price of semiconductor properties (g, α), so an intermediate coupling optimizes the topological gap. Our $g = 2$ ceiling is the sharp small-g limit of that statement: with $E_Z(2\,\mathrm{T}) = 116\,\mu$eV fixed by $g = 2$, the topological condition $E_Z > \sqrt{\Delta_{ind}^2 + \mu^2}$ cannot be met at all for $\Delta_{ind} \geq 116\,\mu$eV, and maximizing the gap over $(B, \Delta_{ind} \leq \Delta_p(B), \mu)$ under GL suppression yields an interior optimum $\Delta_0 \approx 98\,\mu$eV (main-text Sec. 3.2) — i.e. for silicon electrons the Cole optimum is not merely beneficial but *mandatory*, and it caps the achievable gap at a value disorder then destroys. The hole platform escapes through g_z up to 4.2, which moves the ceiling to $\sim 240\,\mu$eV at 2 T.

7 Quasiparticle-poisoning estimate

The equilibrium quasiparticle fraction of the parent at temperature T is $x_{qp} \simeq \sqrt{2\pi k_B T/\Delta_p}\, e^{-\Delta_p/k_B T}$. Requiring $x_{qp} \leq 10^{-6}$ with the Si:B parent gap *at the operating field* ($\Delta_p(B^*) = 29$–$33\,\mu$eV for the measured-field design) gives $T_{\mathrm{eff}} \leq 25$–29 mK; the same criterion for the Al reference point of `qp_poisoning.py` ($\Delta_p = 150\,\mu$eV at $B = 1.0$ T, GL) gives 130 mK. Since realistic electron temperatures in superconducting devices rarely reach 25 mK, the measured-field all-silicon stack is

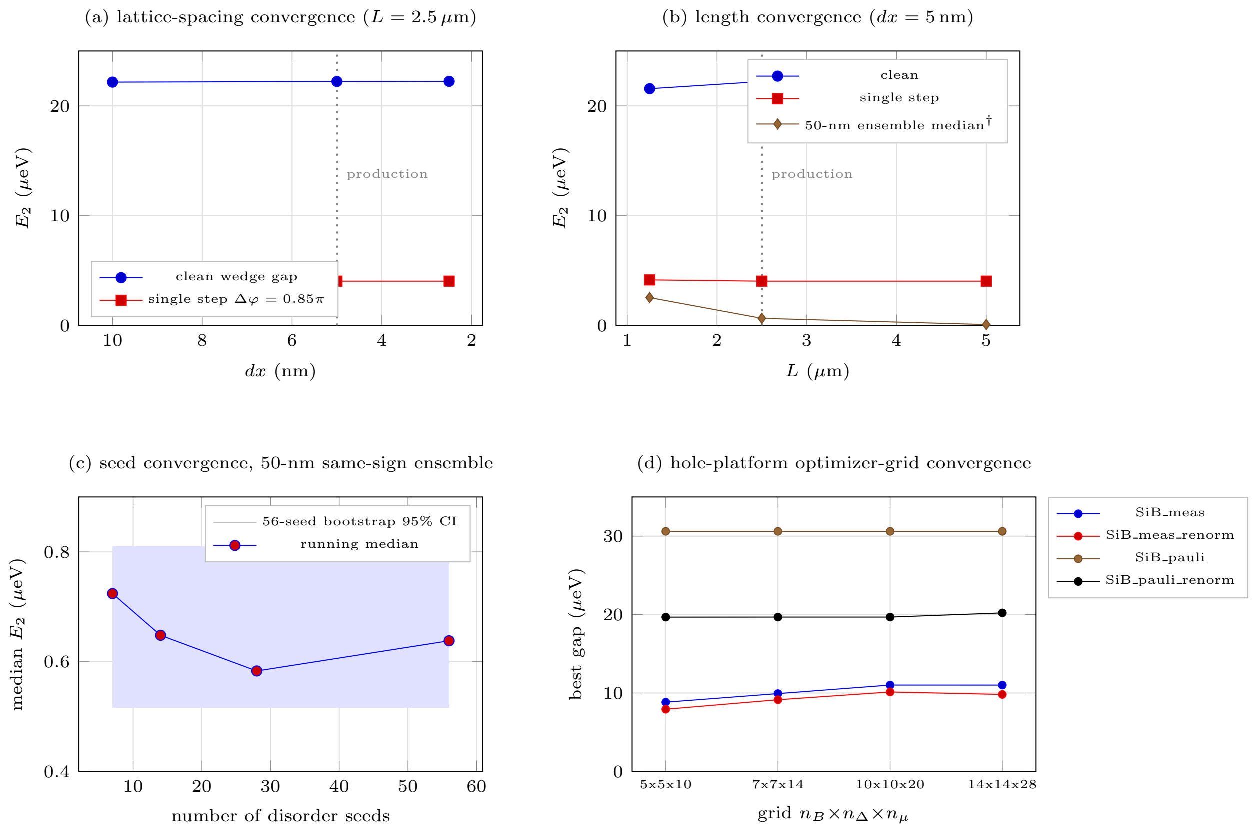

Figure 1: Convergence summary (data: `convergence.json`; rendered by `tools/fig13_pgf.py`, equivalently `convergence.py` with matplotlib). †The L-dependence of the step-ensemble median in panel (b) is physical (Anderson localization), not a discretization artifact; see Table 5, note *a*.

classified as a detection platform (ZBP spectroscopy, gap hardness, the miscut test) rather than a parity-coherent qubit, absent parent-gap engineering. All numbers in `qp_poisoning.py`.

8 Luttinger–Kohn extraction

The fin cross-section (y, z) is solved in the 4-band LK Hamiltonian (hard-wall, grid 11×13, $\gamma_1 = 4.285$, $\gamma_2 = 0.339$, $\gamma_3 = 1.446$, $\kappa = -0.42$) with vertical field E_z; the lowest Kramers doublet is followed in k_x to extract m^* (curvature), α (linear-in-k splitting), the SOC axis (spin direction of the split states), and the g-tensor (bare $2\kappa\mu_B \mathbf{B} \cdot \mathbf{J}$ plus orbital terms, evaluated as the field-derivative of the doublet splitting along the three axes). Validation: at $E_z = 0$ the model reproduces the exact bulk [100] heavy/light masses, and the predicted spin–orbit lengths $l_{so} = \hbar^2/(m^*\alpha) = 32$–$101$ nm overlap the measured FinFET window ($\sim$20–60 nm). Limits: no strain, no cubic (J^3) Zeeman, no device electrostatics — the known shortfall is $g_x \in [0.4, 1.1]$ across a seven-geometry bracket (7–16 nm sections, 3–30 MV/m) versus measured $g_{xx} \approx 1.9$–2.3; all platform numbers are therefore carried under both tensors, and the 6-band+Poisson resolution is the top open theory item.

9 Reproducibility

Every figure and quoted number regenerates from the committed code: `run_analysis.py` (figures 1–11), `transport.py` / `transport_valley.py` (RGF, invariant, valley transport), `lk_holes.py`, `qp_poisoning.py`, `convergence.py` (this supplement's Sec. 5), `tests.py` (assertion suite covering the equivalence theorem, gauge covariance, analytic dispersions, RGF agreement with closed-chain spectra to 10^{-13}, and regression tests for every bug found in review). The numerical ledger is `output/data/key_numbers.json`; scan checkpoints allow resuming every long run. Archival packaging (commit hash, environment, per-figure commands) is described in `README.md`; randomness is fully seeded (seed families are listed in Tables 2–4).